UNIVERSITY OF CENTRAL PUNJAB

(Faculty of IT)

Computer Communications and Networks (CCN)

Assignment No 4 **Date : 09 Jun 2026**

Total Marks: 100 **Due Date : 17 Jun 2026**

CLO3 Questions

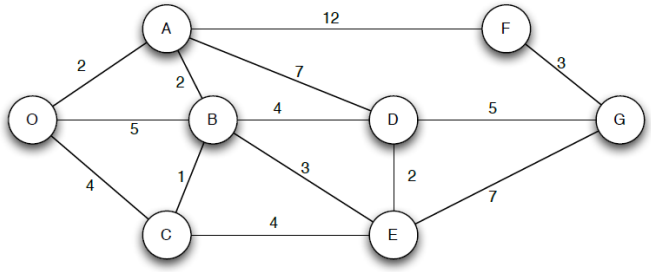
1. Analyze the router behavior when it has to send one 5200 byte IP datagram into a link that has an MTU of 1500 bytes. Consider original datagram is stamped with the identification number 777. How many fragments will be generated, and how much data will be in each fragment. In each of the IP fragments generated, give the values in various fields of the IP headers.

2. A small ISP has the IPv4 address range defined by the prefix 51.134.215.0/24. The ISP has four clients, each requires up to 58 IP addresses. Organize the allocation of IP addresses into four subnets, one for each client; give the subnetwork address of each subnet; also how many IP addresses will be available in each subnet.

3. Suppose an ISP owns the block of addresses of the form 158.47.96.0/19. Suppose it wants to create eight subnets from this block, with each block having the same number of IP addresses. Organize the subnetwork address for each subnet; also mention the broadcast address for each subnet, along with the subnet mask.

4. Consider a router that interconnects three subnets; Subnet 1, Subnet 2 & Subnet 3. Suppose all interfaces in each of these three subnets are required to have the prefix 95.68.55.0/24. Also suppose that Subnet 1 is required to support at least 27 hosts, Subnet 2 is to support at least 83 hosts, and Subnet 3 is to support at least 11 hosts. Organize three network addresses of the form a.b.c.d/x by subnetting, that satisfy these constraints. Show detailed working. Also mention the starting and ending IP address for each subnet, along with the subnet masks. What is the max number of hosts that may be connected to each subnet.

5. Examine the given network, and use Djikstra's shortest-path algorithm with indicated link costs, to prepare the complete routing table for node 'O'.
   Show all computations using relevant equations, and also derive the forwarding table.



Assignment Guidelines:

*Assignment is to be **handwritten**, done in **groups of 4 students**, and submitted in **hard copy**.*

Assignment should have a cover page; should mention your Names, IDs, Section and Assignment number.

*Assignments will **NOT be accepted after the due date.***